

```
apt-get install mercurial
apt-get install qt4-dev-tools
```

If your operating system is Fedora, then type the following commands in the terminal:

```
yum install mercurial
yum install qt4
yum install qt4-devel
```

The above commands will take care of the QT4 development package installation. Now it is time to install NetAnim. If you have followed the installation instructions provided in the previous parts of this series, then you will have a directory named *ns/ns-allinone-3.22/netanim-3.105*. Open a terminal in this directory and execute the following command:

```
make clean
qmake NetAnim.pro
make
```

The command 'qmake NetAnim.pro' might give you the following error message 'qmake: command not found'. The utility called *qmake* automates the generation of Makefiles. *qmake* is not supported in some systems, in which case, execute the command 'qmake-qt4 NetAnim.pro' instead of 'qmake NetAnim.pro' and then execute the command 'make'. This will finish the installation of NetAnim in your system, and you can see the ELF file (executable file format in Linux similar to an exe file in Windows) of NetAnim in the directory *ns/ns-allinone-3.22/netanim-3.105*. Now, you can run an instance of NetAnim by executing the command './NetAnim' in a terminal opened in this directory. But we still don't have an XML based animation trace file which will be used by NetAnim to show us the topology animation. So it is time to go through another ns-3 script, which will give us the XML file to feed NetAnim.

The script for animation and tracing

The ns-3 script named *netanim1.cc* given below will generate an animation trace file with the extension '.xml' and an ASCII trace file with extension '.tr'. You can download the file *netanim1.cc* and all the other script files discussed in this article from sourceforge.net/projects/efytimes/files/2015/08/01/ns3.zip. Running the ns-3 script is similar to what we did before with just the names changed.

```
#include <iostream>
#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
#include "ns3/point-to-point-module.h"
#include "ns3/netanim-module.h"
```

```
#include "ns3/applications-module.h"
#include "ns3/point-to-point-layout-module.h"
```

```
using namespace ns3;
using namespace std;
```

```
int main(int argc, char *argv[])
{
    Config::SetDefault("ns3::OnOffApplication::PacketSize",
    UIntegerValue(1024));
    Config::SetDefault("ns3::OnOffApplication::DataRate",
    StringValue("100kb/s"));
    std::string animFile = "netanim1.xml";
    PointToPointHelper pointToPointRouter;
    pointToPointRouter.SetDeviceAttribute("DataRate",
    StringValue("10Mbps"));
    pointToPointRouter.SetChannelAttribute("Delay",
    StringValue("1ms"));
    PointToPointHelper pointToPointEndNode;
    pointToPointEndNode.SetDeviceAttribute("DataRate",
    StringValue("10Mbps"));
    pointToPointEndNode.SetChannelAttribute("Delay",
    StringValue("1ms"));
    PointToPointDumbbellHelper d(1, pointToPointEndNode, 1,
    pointToPointEndNode,
    pointToPointRouter);
    InternetStackHelper stack;
    d.InstallStack(stack);
    d.AssignIpv4Addresses(Ipv4AddressHelper("192.168.100.0",
    "255.255.255.0"),
    Ipv4AddressHelper("192.168.100.0",
    "255.255.255.0"),
    Ipv4AddressHelper("192.170.100.0",
    "255.255.255.0"));
    OnOffHelper clientHelper("ns3::UdpSocketFactory",
    Address());
    clientHelper.SetAttribute("OnTime", StringValue("ns3::Unifor
mRandomVariable"));
    clientHelper.SetAttribute("OffTime", StringValue("ns3::Unifo
rmRandomVariable"));
    ApplicationContainer clientApps;
    AddressValue remoteAddress(InetSocketAddress(d.
    GetLeftIpv4Address(0), 5000));
    clientHelper.SetAttribute("Remote", remoteAddress);
    clientApps.Add(clientHelper.Install(d.GetRight(0)));
    clientApps.Start(Seconds(1.0));
    clientApps.Stop(Seconds(10.0));
    d.BoundingBox(1, 1, 25, 25);
    AnimationInterface anim(animFile);
    anim.EnablePacketMetadata();
    Ipv4GlobalRoutingHelper::PopulateRoutingTables();
    AsciiTraceHelper ascii;
    pointToPointEndNode.EnableAsciiAll(ascii.
```